

## ROVER LAB, DOSSIER (K2-18 b mission)

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### # Planet

Name: K2-18 b

Planet type: Sub-Neptune 'Hycean' candidate, thick hydrogen atmosphere over a possible global ocean

Surface temperature:  $\approx -8^\circ\text{C}$  equilibrium estimate (no defined solid surface)

Surface pressure: No solid surface; pressure increases with depth through the atmosphere/ocean layer.

Gravity:  $\approx 1.27\text{ g}$

Atmosphere: Methane confirmed by JWST (2023). A disputed, unconfirmed hint of dimethyl sulfide (DMS) has been reported and is NOT accepted as a biosignature by the wider scientific community.

### # Mission Briefing

K2-18 b is a candidate 'Hycean' world: a deep global ocean, possibly hundreds of kilometres deep, sealed beneath a thick hydrogen-rich atmosphere. A submersible probe, 'Abyss-1', has descended through the haze to the layer where the ocean surface is thought to lie, and collected a sample of strangely tinted liquid. Some scientists have proposed that colour and chemistry at this depth could come from dissolved iron minerals seeping from a rocky core far below; others think any colouration is more likely organic. Mission control needs the sample tested before deciding whether this world is worth the huge energy cost of a full mining mission.

### # Rover

Starting energy: 100

Site visit cost: 10

Experiment cost: 15

### # Overview Image

### # Site: Tinted liquid from the upper ocean layer

Description: A sample of pale amber-tinted liquid drawn from just beneath the cloud deck, in permanent twilight far below the visible cloud tops.

Identity: Food-colourant solution (amber dye in water), no real iron content. DECOY GROUP.

### ## Known Properties

Colour: Pale amber

Odour: None

Solubility: Fully miscible with water

### ## Experiment Results

Measure pH: Close to neutral, pH around 7

Measure density: 1.0 g/mL

Test electrical conductivity: Low

Add a magnet: No attraction

Add potassium thiocyanate solution: No colour change

Add sodium hydroxide solution: No precipitate forms

Heat a small sample gently: No visible change

### # Experiments

Measure pH

Measure density

Test electrical conductivity

Add a magnet

Add potassium thiocyanate solution

Add sodium hydroxide solution

Heat a small sample gently

### # AI Behaviour

Reveal remaining candidates after narrowing: No

Socratic questions: Yes

Difficulty: Standard (15-16 years old)

## # Teacher Notes

DECOY GROUP, assign a food-colourant solution, not real  $\text{FeCl}_3$ . K2-18b is real and famous for a genuinely disputed claim (dimethyl sulfide, a proposed biosignature), a great planet to pair with a fake sample, because it lets you make the point explicit: an exciting, headline-making claim about a real planet doesn't mean every 'sample' from it is what it appears to be. Use the final conference to connect this directly to how real scientists must independently verify extraordinary claims before accepting them (see DMS controversy below), the fake sample is a hands-on echo of that same lesson.

Real-science notes: K2-18b is a genuine sub-Neptune,  $\sim 124$  light-years away, orbiting the red dwarf K2-18 every 32.9 days, discovered in 2015. JWST (2023, Madhusudhan et al.) robustly detected methane in its atmosphere, a real, solid measurement. A 2023 hint and 2025 follow-up claim of dimethyl sulfide (DMS), proposed as a possible biosignature, has been strongly disputed: multiple independent reanalyses in 2025 found the statistical evidence insufficient, and DMS can also form without life. Its low density ( $2.67 \text{ g/cm}^3$ , about half Earth's) means it is NOT a rocky terrestrial planet, the 'Hycean ocean world' picture is one leading hypothesis, but an ordinary gas-rich mini-Neptune with no ocean at all is an equally consistent alternative model. Teach this planet as a case study in scientific caution: methane = solid finding; DMS-as-life-signal = not yet accepted.

Sources: Madhusudhan et al. 2023 ([science.org/content/article/alien-planet-s-atmosphere-bears-chemical-hints-life-astronomers-claim](https://science.org/content/article/alien-planet-s-atmosphere-bears-chemical-hints-life-astronomers-claim)); 2025 reanalysis, A&A ([aanda.org/articles/aa/full\\_html/2025/08/aa55580-25/aa55580-25.html](https://aanda.org/articles/aa/full_html/2025/08/aa55580-25/aa55580-25.html)).

Cross-subject link: Biology selective pressure for this planet, no sunlight reaches the deep ocean floor beneath a thick hydrogen atmosphere; organisms would need an energy source other than photosynthesis (chemosynthesis, as in Earth's deep-sea hydrothermal vent life).